

**OCEANSENSE: AN IOT-BASED HYBRID SYSTEM FOR
REAL-TIME DETECTION AND TRACKING OF
MARINE DEBRIS**

25-26J-002

Project Proposal Report

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BSc (Hons) in Information Technology Specializing in Computer
Systems & Network Engineering

Department of Computer Systems Engineering

Sri Lanka Institute of Information Technology
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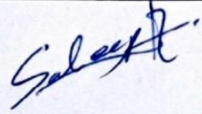
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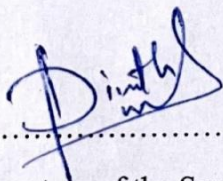
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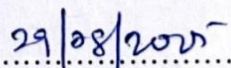
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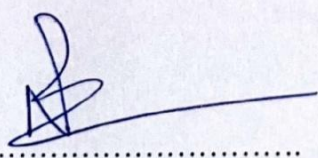
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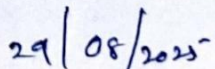
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ABSTRACT

The escalating crisis of marine plastic pollution necessitates the development of autonomous, energy-resilient monitoring systems capable of operating reliably in harsh and dynamic ocean-surface environments. OceanSense addresses this need through a distributed IoT swarm architecture designed for real-time detection and tracking of floating marine debris. This proposal specifically presents the design, implementation, and validation of the Energy-Resilient Adaptive Communication (ERAC) protocol—a custom Medium Access Control (MAC) framework optimized for energy-constrained buoyant sensor nodes communicating using point-to-point Long Range (LoRa) links.

Marine surface IoT systems face several unique challenges, including highly stochastic energy availability from solar and wave sources, intermittent connectivity caused by wave-induced occlusion, and the need for strict Energy Neutral Operation (ENO) to ensure long-term autonomy. ERAC addresses these constraints using a cross-layer design that tightly integrates lightweight energy prediction with QoS-aware packet scheduling. By combining EWMA-based solar forecasting and inertial-based wave energy estimation, ERAC dynamically adapts node duty cycles and transmission parameters (Spreading Factor, Transmit Power) to prioritize critical debris alerts while deferring routine telemetry during energy-limited periods.

A dual-track validation methodology is adopted. High-fidelity NS-3 simulations evaluate scalability, packet reliability, and contention behavior under realistic marine channel models for networks of 10–15 nodes. In parallel, a hardware prototype consisting of an ESP32/Raspberry Pi-based master node and sub-nodes is used to validate ERAC’s energy-aware scheduling and LoRa link robustness in practice. This report presents theoretical background, design rationale, research gaps, methodology, and requirements, establishing ERAC as a focused and practical contribution to resilient, energy-neutral marine monitoring.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
ADC	Analog-to-Digital Converter
ADR	Adaptive Data Rate
CSS	Chirp Spread Spectrum
DC	Duty Cycle
EH-MAC	Energy Harvesting Media Access Control
EH-WSN	Energy Harvesting Wireless Sensor Networks
ENO	Energy Neutral Operation
ERAC	Energy-Resilient Adaptive Communication
EWMA	Exponentially Weighted Moving Average
IoT	Internet of Things
LoRa	Long Range
LoRaWAN	Long Range Wide Area Network
MAC	Media Access Control
NFR	Non-Functional Requirement
NS-3	Network Simulator 3
PDR	Packet Delivery Ratio
QoS	Quality of Service
SF	Spreading Factor
SO	Specific/Sub Objective
TC	Test Case
T _x	Transmit / Transmission
TDMA	Time Division Multiple Access
UAV	Unmanned Aerial Vehicle
UWSN	Underwater Wireless Sensor Network
WEC	Wave Energy Converter

** All abbreviations are defined at first use in the text.*

1 INTRODUCTION

1.1 Global Context: Marine Debris and the Critical Need for UWSN Monitoring:

The accumulation of anthropogenic debris, specifically plastic pollution, in the world's oceans has precipitated a global environmental crisis of unprecedented scale. Estimates suggest that millions of metric tons of plastic enter the marine environment annually, persisting for centuries and fragmenting into microplastics that infiltrate the entire marine food web. The ecological ramifications are severe, ranging from the entanglement and ingestion of debris by marine life to the chemical contamination of habitats. Beyond ecological impacts, marine debris imposes substantial economic burdens on coastal communities, tourism industries, and maritime navigation.

Current monitoring methodologies remain heavily reliant on sporadic surveys, satellite imagery, and aerial surveillance. While remote sensing technologies utilizing satellites and Unmanned Aerial Vehicles (UAVs) have advanced the capability to detect large aggregations of floating debris, they are often constrained by temporal resolution, cloud cover, and significant operational costs. Satellites, for example, may only capture a specific coastal region once every few days, potentially missing rapid debris flows driven by storm surges or tidal changes. Furthermore, these high-altitude observations often lack the granularity required to detect smaller, dispersed debris fields that precede the formation of massive garbage patches. [1] [2]

To overcome these limitations, the OceanSense project proposes a transition towards continuous, in-situ monitoring using a distributed network of autonomous sensor nodes. By deploying a swarm of low-cost, buoyant IoT devices directly into the water, OceanSense aims to provide real-time, high-resolution data on debris presence and drift patterns. This approach demands a rethinking of marine communication infrastructures, moving away from expensive, power-hungry instrument buoys toward fleets of energy-neutral, intelligent sensor nodes capable of forming a resilient observation network.

1.2 Problem Domain: Challenges in Dynamic, Energy-Constrained UWSNs:

Deploying Wireless Sensor Networks (WSNs) in the marine surface layer introduces a unique and formidable set of challenges that render traditional terrestrial networking protocols ineffective. The primary constraint governing the design of such systems is severe energy scarcity. Marine buoys must operate autonomously for extended periods and rely entirely on energy harvesting (EH) to sustain operations. OceanSense uses a hybrid harvesting strategy – solar photovoltaic (PV) panels and wave-induced kinetic harvesters – both of which are inherently stochastic. Solar energy follows a diurnal pattern but is affected by cloud cover and biofouling, while wave energy fluctuates with sea state, wind speed, and swell direction. [3] [4] [5] [6]

The second critical challenge is the communication channel itself. OceanSense employs LoRa (Long Range) modulation technology, operating in the sub-GHz ISM bands (e.g., 868 MHz or 915 MHz), to achieve long-range communication with minimal power consumption. While LoRa's Chirp Spread Spectrum (CSS) modulation offers excellent sensitivity and resistance to interference, its application in the marine environment is complicated by the "near-surface" propagation effect. As buoys heave and pitch with the waves, the Fresnel zone is frequently obstructed by the water surface, leading to rapid fluctuations in signal strength and intermittent connectivity – a phenomenon known as wave occlusion. Furthermore, standard LoRaWAN protocols, which rely on ALOHA-based random access, suffer from scalability issues. As the number of nodes increases, the probability of packet collisions rises exponentially, leading to data loss and energy-wasting retransmissions. [7] [8] [9] [10]

Furthermore, debris-monitoring traffic is inherently bursty and priority-driven. Under normal conditions, nodes transmit low-frequency “heartbeat” packets. Upon detecting debris, however, nodes generate latency-critical alerts that must reach the coordinator promptly. Standard, generic MAC protocols that treat all packets equally cannot ensure the reliable and timely delivery of such high-value data during congestion or low-energy conditions [11] [12]. These constraints necessitate a specialized, energy-aware communication framework

1.3 Overview of the OceanSense Hybrid System

OceanSense is designed as a modular, scalable IoT system for near-surface marine monitoring. The system comprises a cluster of Sensor Nodes (Sub-Nodes) and a central Coordinator Node (Master Node).

- **Sensor Nodes:** These are compact, buoyant units equipped with a microcontroller (e.g., ESP32-S3, STM32WL or Raspberry Pi), a LoRa transceiver (SX1276/SX1262), and a sensor suite including optical cameras for visual debris identification and Inertial Measurement Units (IMUs) for wave parameter estimation. They harvest energy via top-mounted solar panels and internal triboelectric or electromagnetic wave energy converters (WECs). [4] [13]
- **Master/Coordinator Node:** This node serves as the sink for the swarm, aggregating data from the sub-nodes and forwarding it to a cloud backend via a high-bandwidth backhaul (e.g., cellular LTE/5G or Satellite IoT). It manages network synchronization and broadcasts global parameters.
- **Communication Model:** The system employs direct point-to-point LoRa links, not full LoRaWAN. This design choice grants complete control over timing, duty-cycling, and scheduling—essential for ERAC’s energy-aware and priority-aware behavior.

The entire system adheres to the principle of Energy Neutral Operation (ENO): across any operational window, total energy consumed must not exceed energy harvested. This ensures long-term, maintenance-free operation of deployed nodes. [14] [15]

1.4 Focusing the Component: Energy-Aware MAC & Network Resilience

This proposal specifically addresses the design and validation of the Energy-Resilient Adaptive Communication (ERAC) protocol. ERAC forms the intelligent MAC layer that sits between the application layer and the LoRa physical layer. Its mandate is to orchestrate channel access and transmission behavior to maximize reliability and ensure longevity under fluctuating energy constraints.

ERAC distinguishes itself from conventional LoRa-based MAC approaches – often analyzed in the context of LoRaWAN literature – through two core mechanisms:

1. Predictive Energy-Awareness:

ERAC employs solar and wave energy forecasting to anticipate future energy availability. The protocol dynamically adjusts node duty cycles, conserving energy during predicted shortages and utilizing surplus periods to improve sensing and transmission fidelity. [5] [16]

2. QoS-Driven Scheduling:

ERAC implements a priority-based queuing mechanism that differentiates between critical debris alerts and routine telemetry. By adjusting LoRa transmission parameters such as Spreading Factor (SF) and backoff intervals according to packet priority, ERAC ensures that high-value information receives preferential channel access. [17] [18]

This contribution focuses exclusively on the MAC and energy-interaction layers, intentionally excluding broader aspects such as swarm coordination or underwater optical communication, to deliver a rigorous, optimized solution to the energy-connectivity trade-off in marine IoT systems.

2 BACKGROUND AND LITERATURE REVIEW

2.1 Evolution of MAC Protocols in Underwater Wireless Sensor Networks (UWSNs)

Traditional Medium Access Control (MAC) protocols for marine and underwater environments have historically focused on acoustic communication, given the severe attenuation of radio signals below the surface. Classic protocols such as FAMA and MACA-U employ RTS/CTS-style coordination to mitigate collisions under long propagation delays. However, the high cost and power requirements of acoustic modems make them unsuitable for low-cost, energy-constrained deployments like OceanSense.

Consequently, recent research has shifted toward adapting terrestrial Low-Power Wide-Area Network (LPWAN) technologies – especially LoRa – for near-surface marine monitoring. LoRa’s Chirp Spread Spectrum (CSS) modulation provides excellent link budget and robustness with minimal transmission power, making it advantageous for surface-deployed buoys.

Much of the literature frames LoRa behavior through LoRaWAN, whose Class A devices employ pure ALOHA transmissions. Studies highlight that ALOHA-based access suffers from severe scalability issues; as node density increases, collisions become a dominant factor, resulting in substantial Packet Delivery Ratio (PDR) degradation. The challenge intensifies in marine environments, where “hidden nodes” emerge due to wave-induced line-of-sight blockages.

Although proposals such as slotted ALOHA or TDMA-over-LoRa attempt to introduce determinism, they require precise network-wide synchronization. Maintaining such timing on energy-harvesting buoy nodes incurs heavy energy overhead. As a result, current trends emphasize asynchronous and adaptive MAC designs that rely on local intelligence – rather than global synchronization – to manage channel access efficiently.

This evolutionary direction motivates the ERAC protocol, which builds upon LoRa’s PHY-layer flexibility but replaces rigid LoRaWAN-style MAC behavior with a custom, energy-aware, priority-driven scheduler suitable for marine surface IoT.

[19] [20]

2.2 State-of-the-Art in Energy Harvesting MAC (EH-MAC) Protocols

Energy Harvesting MAC (EH-MAC) protocols redefine the classical objective of lifetime maximization. Rather than slowing the depletion of a finite battery, modern EH-MACs aim to achieve Energy Neutral Operation (ENO), where average harvested energy matches or exceeds energy consumption. [13] [21]

Early EH-MAC designs used simple reactive strategies—nodes reduced activity only when battery level dropped below a threshold. However, these approaches prove insufficient in environments with highly variable harvesting conditions, such as marine surfaces where sunlight and wave activity fluctuate unpredictably.

Recent literature emphasizes prediction-based MAC strategies, in which the node anticipates future energy income and adjusts its operations proactively. This shift is essential for proper scheduling of sensing, computation, and communication tasks in low-power IoT systems.

2.2.1 Predictive Energy Management and Adaptive Duty Cycling

Adaptive Duty Cycling is central to EH-MAC design. Instead of fixed wake-up intervals, the MAC layer adjusts node activity based on predicted energy availability.

- **Solar Prediction:** Algorithms like the Exponentially Weighted Moving Average (EWMA) have been refined to include diurnal (daily) corrections, improving accuracy for solar harvesting. More advanced methods employing lightweight Machine Learning (ML), such as TinyML implementations of regression models or even simplified Long Short-Term Memory (LSTM) networks, have been demonstrated on edge devices. These models allow nodes to anticipate energy "income" and plan their "expenditure" (transmissions) accordingly. [6] [22]
- **Wave Energy:** Predicting wave energy is more complex due to its irregular stochastic nature compared to the solar cycle. Research by Deberneh and Kim utilizes inertial data (accelerometers) processed via lightweight neural networks or regression analysis to forecast near-term wave energy potential. Integrating these dual-source predictions into a single MAC decision engine remains a cutting-edge challenge, as discussed in [23], which proposes a hybrid solar-vibration prediction model for WSNs.

2.2.2 Cross-Layer Integration in EH-WSNs

To be truly effective, energy awareness cannot be siloed in the physical layer. Cross-layer protocol designs, where the MAC layer has direct visibility into the energy subsystem's State of Charge (SoC) and predicted harvesting rate, have demonstrated superior performance. Protocols such as those reviewed in [24] utilize this cross-layer information to dynamically adjust backoff windows and retransmission limits. For instance, if the energy prediction is bleak, the MAC layer might aggressively aggregate data or increase sleep intervals, a behavior impossible in a strictly layered OSI model. The ERAC protocol builds upon this by creating a direct feedback loop between the energy harvester and the LoRa transmission scheduler.

2.3 Literature on QoS Provisioning in UWSNs

Quality of Service (QoS) support is critical in mission-driven networks but remains a developing area in LoRa-based systems. Standard LoRaWAN MAC behavior treats all packets as best effort, lacking innate priority differentiation. This limitation is problematic for debris-monitoring systems where critical alerts must supersede routine telemetry.

2.3.1 Coupling Energy Harvesting and QoS Requirements

Recent studies have proposed various mechanisms to inject QoS into LoRa / LoRaWAN networks.

- **Priority Queuing:** Implementing multiple queues at the node level, served by strict priority or weighted round-robin scheduling.
- **Parameter Manipulation:** Assigning robust Spreading Factors (e.g., SF12) to high-priority packets to ensure reachability or using lower SFs for low-priority data to save airtime. [25]
- **Reinforcement Learning (RL):** Using RL agents to learn the optimal transmission parameters for different traffic classes under varying channel conditions. [17] [26]

However, a critical gap identified in the literature is the Energy-QoS trade-off. High reliability (high QoS) typically demands high energy (higher transmission power, more retransmissions). In an energy-harvesting context, satisfying QoS requirements might lead to energy depletion and node death. Work by Khan et al. (2022) begins to address this by proposing "energy-aware QoS" where the QoS target is dynamically scaled based on energy reserves. ERAC adopts this philosophy, introducing a mechanism where the "cost" of a transmission is weighed against its "priority" and the "available budget". [18]

2.4 Network Fault Tolerance and Resilience in LoRa-Based Marine IoT

Marine environments exhibit highly variable channel quality due to buoy motion, wave occlusion, and dynamic range variation. Ensuring resilient communication requires adaptive link management and congestion-aware MAC logic.

2.4.1 Adaptive Data Rate and Link Robustness

The marine channel is characterized by rapid fading and path loss fluctuations. Standard LoRaWAN Adaptive Data Rate (ADR) algorithms, which average Signal-to-Noise Ratio (SNR) over 20 packets, are often too slow to react to marine channel dynamics. This lag leads to packet loss during sudden fades (e.g., a high wave blocking the signal). Advanced algorithms like "Blind ADR" or collision-aware ADR have been proposed to improve responsiveness. For marine applications, maintaining connectivity is often prioritized over maximizing data rate; therefore, robust, conservative rate selection strategies are favored. [27] [28] [29]

These studies inform ERAC's fast-drop, slow-rise strategy: when rapid degradation is detected, the MAC immediately switches to a more robust SF, maintaining connectivity during brief occlusions.

2.4.2 MAC-Level Connectivity Maintenance

Fault tolerance also involves handling congestion. When multiple nodes detect debris simultaneously (a spatially correlated event), they may flood the channel, causing a collision cascade. Techniques such as Distributed Queuing or adaptive backoff algorithms are essential to desynchronize these transmissions. ERAC integrates a priority-based backoff mechanism, ensuring that even in a congested scenario, the most critical alerts have the highest probability of seizing the channel. [30]

3 RESEARCH GAP ANALYSIS:

The review of existing marine IoT, LoRa-based, and EH-MAC literature reveals several critical shortcomings that directly motivate the development of the Energy-Resilient Adaptive Communication (ERAC) protocol. While significant progress has been made in energy harvesting systems and adaptive MAC behavior, important gaps remain – particularly in the context of surface-level marine debris monitoring, where stochastic energy availability and bursty, priority-driven traffic must be balanced within an ultra-low-power LoRa communication framework.

3.1 Gap in Unified Cross-Layer EH-MAC Design for Hybrid UWSNs

Current research commonly treats Energy Harvesting (EH) and MAC scheduling as largely separate optimization domains. In many LoRa- or LoRaWAN-focused studies, MAC algorithms are designed without direct access to predicted harvesting rates or future energy estimates. Conversely, energy prediction techniques – such as solar EWMA, TinyML forecasting, and wave-energy inference – are often evaluated in isolation from the communication layer and not integrated with transmission scheduling decisions.

This disconnect is particularly problematic in a marine context, where both solar and wave energy sources exhibit rapid fluctuations. Existing systems tend to either:

- over-transmit under momentary surplus, causing future depletion, or
- excessively conserve energy, under-utilizing predictable surpluses.

Gap Identified:

There is no unified, cross-layer framework that integrates dual-source stochastic energy prediction directly into the MAC scheduling logic for LoRa-based marine nodes. Existing designs either neglect the dynamic nature of dual-source EH or fail to embed predictive intelligence into MAC decisions, leading to suboptimal duty cycling and unreliable data delivery.

3.2 Deficiency in Resilient Networking under Swarm Mobility

Most LoRa and LoRaWAN MAC techniques are evaluated for static terrestrial deployments – smart meters, agriculture sensors, and building monitoring. In contrast, surface-deployed buoys drift with currents, pitch with waves, and experience frequent shadowing due to water-surface occlusion. These dynamics introduce:

- rapid, short-term SNR fluctuations,
- intermittent link outages, and
- hidden-node scenarios caused by wave crests.

Standard adaptive data rate mechanisms (e.g., slow ADR averaging) fail to react at the required temporal scale. Similarly, many QoS approaches do not consider energy state, resulting in nodes exhausting their power reserves during high-traffic situations. [29]

Gap Identified:

There is a lack of resilient, mobility-aware MAC strategies that can rapidly adjust LoRa transmission parameters—such as spreading factor, power, and backoff—based on link fluctuations and predicted energy conditions. Existing algorithms do not jointly optimize resilience, latency, and energy neutrality for marine drift environments.

3.3 Lack of Practical Validation for Stochastic Energy Forecasting

Although many studies propose advanced prediction techniques for solar and wave energy, most rely on synthetic datasets, desktop-grade simulations, or computationally heavy models unsuitable for microcontroller deployment. Lightweight approaches such as EWMA and regression-based wave-state estimation have been suggested but rarely validated on real embedded hardware.

Furthermore, there is limited research demonstrating:

- how prediction accuracy impacts MAC performance,
- whether lightweight models can run on constrained nodes like ESP32/Raspberry Pi,
- and how forecast errors influence duty cycling and QoS scheduling in real time.

Gap Identified:

There is a shortage of practical, hardware-level validation showing that lightweight energy forecasting algorithms can be executed on low-power MCUs and meaningfully guide MAC behavior in stochastic marine environments. Most existing work does not bridge the gap between theoretical accuracy and operational feasibility.

4 RESEARCH PROBLEM:

The deployment of marine surface sensor networks for debris monitoring is constrained by highly variable energy availability and intermittent LoRa connectivity caused by wave occlusion, mobility, and environmental instability. Existing LoRa-based MAC protocols do not simultaneously guarantee energy neutrality and priority-driven delivery of critical event data under these conditions. Standard LoRaWAN mechanisms, designed for static terrestrial devices, lack the adaptive responsiveness, cross-layer energy awareness, and fine-grained QoS differentiation needed for drifting, energy-harvesting marine buoys.

Research Problem:

How can a novel cross-layer Energy-Resilient Adaptive Communication (ERAC) protocol be designed and implemented to enable reliable, priority-based data transmission for marine debris monitoring, while ensuring energy neutrality through predictive management of stochastic solar and wave energy sources on resource-constrained LoRa nodes?

5 OBJECTIVES

5.1 Main Objective:

To design, develop, and validate the Energy-Resilient Adaptive Communication (ERAC) protocol – a cross-layer, energy-aware MAC framework for point-to-point LoRa-based marine IoT networks – that integrates predictive dual-source energy harvesting with QoS-driven packet scheduling to ensure long-term autonomous operation and reliable debris detection alerting.

5.2 Sub-Objectives:

- **SO 1 (ERAC Architecture Design):**
To design a cross-layer MAC architecture that interfaces the physical energy harvesting hardware with the data link layer, enabling dynamic access to State-of-Charge (SoC) and energy generation rates for protocol decision-making.
- **SO 2 (Predictive Energy Management):**
To develop and implement lightweight algorithms for forecasting solar and wave energy availability on embedded microcontrollers (EWMA-based), enabling adaptive duty cycling that prevents energy depletion while maximizing network availability.
- **SO 3 (QoS-Aware Scheduling):**
To design a priority-based packet scheduling mechanism within the MAC layer that differentiates between critical debris alerts and routine telemetry, optimizing transmission parameters (Spreading Factor, backoff timers) to guarantee delivery of high-priority data.
- **SO 4 (Simulation Validation):**
To validate the scalability and reliability of ERAC under varying node densities (10–15 nodes) and stochastic marine channel conditions using the NS-3 network simulator.
- **SO5 (Hardware Implementation):**
To implement the ERAC protocol on a physical prototype consisting of a master node and two sub-nodes, demonstrating the real-world feasibility of energy-aware duty cycling and priority scheduling.

5.3 Statement of Novelty

ERAC introduces a novel dual-source predictive MAC scheduler tailored for the marine IoT domain. Unlike generic EH-MACs that focus solely on solar energy, ERAC explicitly models and predicts the complementary nature of solar and wave energy to optimize duty cycles. Furthermore, it integrates this energy intelligence with a QoS-aware LoRa transmission scheduler, forming a unique cross-layer loop in which the “urgency” of a packet negotiates with the node’s “future energy wealth.”

This combined prediction–QoS–resilience integration, validated through both high-fidelity simulation and hardware-level implementation, addresses a distinct gap in existing literature by enabling resilient, energy-neutral communication for long-term marine debris monitoring.

6 METHODOLOGY

The development of the Energy-Resilient Adaptive Communication (ERAC) protocol follows a dual-track methodology comprising (1) high-fidelity simulation for large-scale validation and (2) real hardware implementation for practical feasibility verification. This combined approach ensures that ERAC is both theoretically robust and deployable on the OceanSense prototype.

6.1 Overall System Description and Component Integration:

The OceanSense architecture consists of a swarm of buoyant sensor nodes (sub-nodes) communicating with a central gateway (master node) via direct point-to-point LoRa links. This communication model is intentionally selected instead of full LoRaWAN to allow fine-grained control over duty cycling, backoff behavior, transmission scheduling, and resilience mechanisms.

- **Sub-Nodes:**
Each sub-node is equipped with sensors (camera, IMU), hybrid energy harvesters (solar + wave), and a LoRa transceiver (SX1276/SX1262). Sub-nodes handle local sensing, lightweight debris detection, and adaptive transmission based on ERAC logic.
- **Master Node:**
The master node collects data from the swarm and forwards it to a cloud backend over an LTE/5G or satellite link. It also periodically broadcasts synchronization beacons or network configuration parameters to assist with loose coordination.

ERAC is embedded within the firmware of each node and interacts closely with the Energy Management Unit (EMU) for real-time feedback on battery state and harvested energy. It also interfaces with the Application Layer to process data packets and priority classifications.

6.2 Hardware and Network Architecture Diagram:

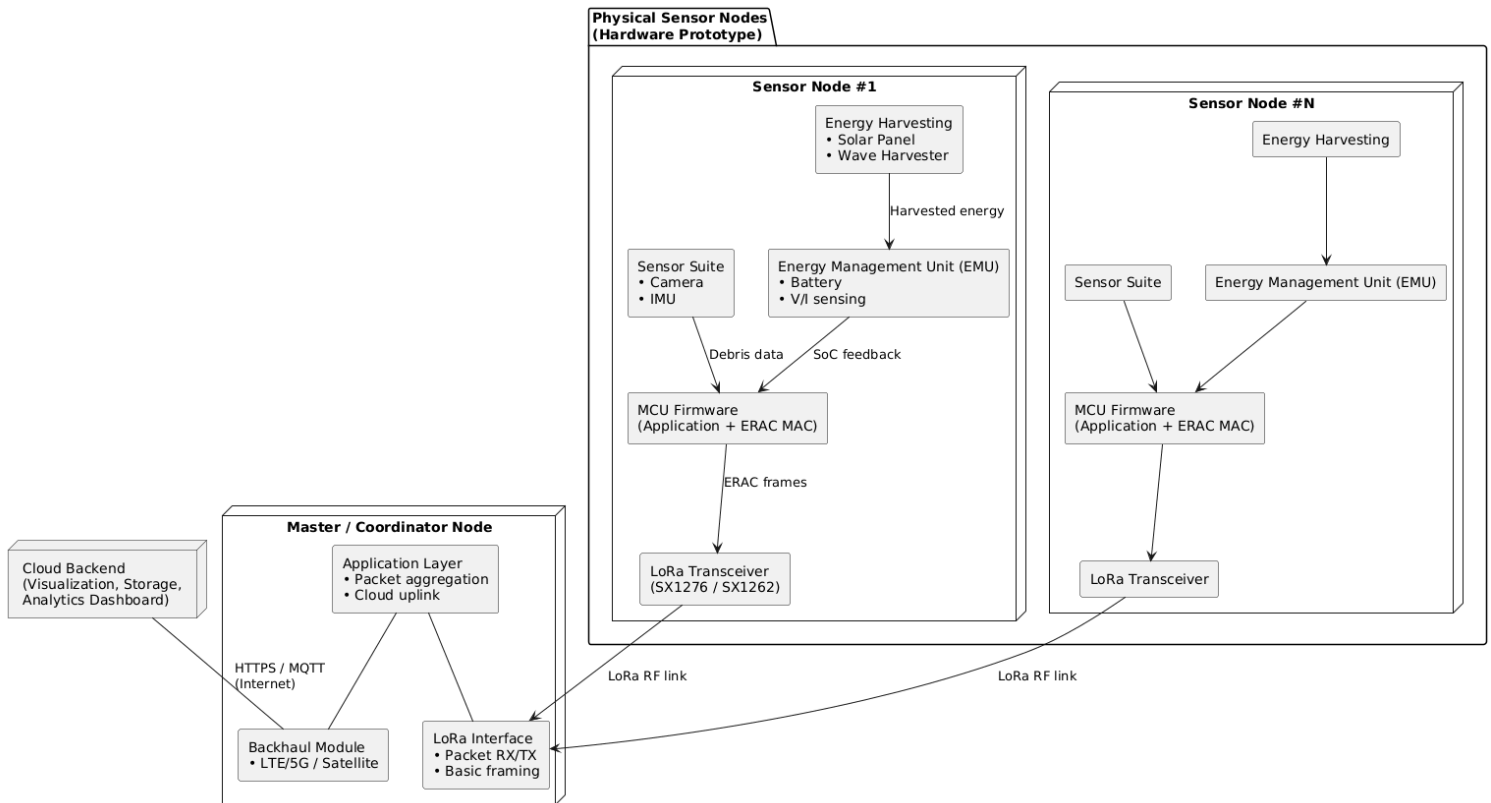


Figure 1 - OceanSense: Hardware & Network Architecture Diagram

6.3 Logical Stack and Simulation Architecture Diagram:

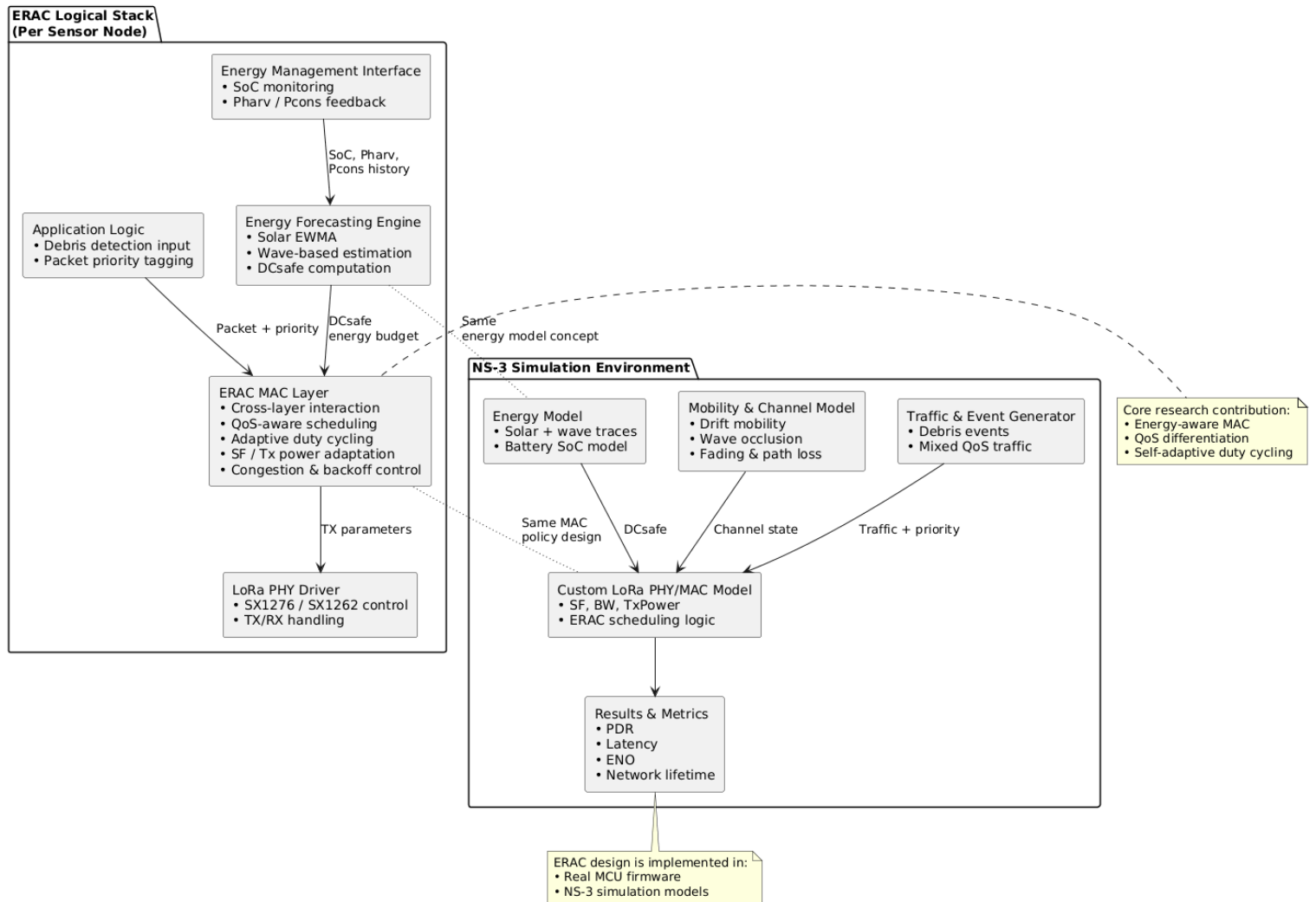


Figure 2 - OceanSense: Logical Stack & Simulation Architecture

6.4 The Energy-Resilient Adaptive Communication (ERAC) Protocol Stack

ERAC is implemented as a cross-layer protocol composed of three tightly integrated modules:

1. Cross-Layer Interaction Framework
2. Energy Forecasting & Adaptive Duty Cycling
3. QoS-Aware Transmission Scheduling

ERAC replaces generic LoRa MAC behavior with an energy- and priority-aware scheduling logic that responds to dynamic marine conditions.

6.4.1 Cross-Layer Interaction Framework

This module enables real-time information exchange between the application, energy subsystem, and MAC layer. ERAC maintains a shared State Register, accessible to both sensing and communication components.

Inputs to ERAC:

- E_{res} (Residual Energy): Current battery voltage read via the ADC.
- P_{harv} (Harvesting Power): Instantaneous current from Solar and Wave sources.
- Q_{len} (Queue Length): Current outgoing queue length.
- $Prio_{pkt}$ (Packet Priority): Tag assigned by the application (1 = Critical, 0 = Routine).

Outputs generated by ERAC:

- DC_{Target} (Target Duty Cycle): The calculated allowable active time per cycle.
- Tx_{params} (Transmission Parameters): Spreading Factor (SF) and Transmit Power (TP).

By coupling energy and traffic information, ERAC avoids “blind” MAC behavior – for example, transmitting low-priority data when the battery is critically low or selecting overly aggressive transmission parameters during energy scarcity. The cross-layer visibility ensures that every packet transmission is judged against the predicted energy budget and system priorities.

6.4.2 Energy Forecasting and Adaptive Duty Cycle Module

This module ensures Energy Neutral Operation (ENO) by forecasting short-term energy availability and adjusting the node's active/sleep cycle accordingly. Marine surface environments exhibit strong temporal variability; therefore, prediction is essential to prevent unexpected depletion.

The fundamental equation Energy Neutrality Condition:

$$\int_0^T P_{cons}(t)dt \leq \int_0^T P_{harv}(t)dt + E_{batt}(0)$$

Where P_{cons} is power consumed and P_{harv} is power harvested over period T .

Prediction Algorithm:

To predict P_{harv} , ERAC employs a lightweight Exponentially Weighted Moving Average (EWMA) with Diurnal Correction.

- Solar Prediction (E_{solar}^{pred}): Uses a slotted history table (e.g., 24 one-hour slots). The prediction for hour h is a weighted average of energy harvested in hour h of previous days and the immediate past hour.

$$E_{pred}(d, h) = \alpha \cdot E_{meas}(d - 1, h) + (1 - \alpha) \cdot E_{meas}(d, h - 1)$$

Where α is a weighting factor (typically 0.7 for historical weight).

- Wave Prediction (E_{wave}^{pred}): Wave energy is less periodic. A simplified linear regression model or a shorter-term moving average is used, utilizing accelerometer data variance as a proxy for sea state. [23]

Duty Cycle Adaptation:

Based on the predicted total energy $E_{total}^{pred} = E_{solar}^{pred} + E_{wave}^{pred}$, the module calculates the Maximum Safe Duty Cycle (DC_{safe}).

- If $E_{res} < E_{crit}$ (Critical Threshold), DC_{safe} is set to the bare minimum for heartbeat (DC_{min}).
- If $E_{res} < E_{high}$ (Surplus), DC_{safe} is increased to allow frequent sensing and transmission.

6.4.3 QoS-Aware Transmission Scheduling (Priority MAC)

This module manages the transmission queue based on DC_{safe} and packet priority.

Traffic Classes:

1. Class 1 (High Priority): Debris Detection Alerts. Latency-critical.
2. Class 2 (Medium Priority): Aggregated Statistics / Wave Data. Delay-tolerant.
3. Class 3 (Low Priority): Heartbeats / Status Logs. High latency tolerance.

Scheduling Logic:

- Priority Queues: Packets are sorted into three distinct queues.
- Energy-Aware Transmission Check: Before extracting a packet from Queue i , the scheduler checks if the energy cost of transmission E_{tx} satisfies:

$$E_{res} - E_{tx} = E_{thresh}(i)$$

Where $E_{thresh}(i)$ is the minimum battery reserve required to send a packet of priority i . For Class 1, this threshold is very low (allow transmission even if battery is low). For Class 3, the threshold is high (only transmit if battery is healthy).

- MAC Parameter Adaptation:
 - High-priority packets → lower SF (fast), higher Tx power
 - Low-priority packets → higher SF only if channel requires it
 - Dynamic adjustment of backoff window based on priority
- Adaptive Backoff: If the channel is busy or collision is inferred, Class 1 packets use a shorter random backoff window compared to Class 3, statistically prioritizing their access to the medium.

This priority-centric design ensures that critical debris notifications “win” channel access during high-traffic events while conserving energy during low-priority transmissions.

6.5 Resilience and Connectivity Maintenance Mechanisms

LoRa communication in marine environments is challenged by fading, wave occlusion, and mobility-induced range variation. ERAC incorporates resilience mechanisms to maintain reliable connectivity without requiring a full LoRaWAN stack.

6.5.1 Adaptive Link Maintenance and Recovery

Marine links fluctuate rapidly as waves obstruct line-of-sight. LoRaWAN's slow ADR (averaging many packets) is unsuitable. ERAC instead uses a fast-drop, slow-rise adaptation:

- On link degradation (missing ACK-equivalent beacon or SNR drop):
→ Immediately increase SF (e.g., SF7 → SF9 → SF11) to maximize link robustness.
- Maintain robust SF for a hold-down period to avoid oscillation.
- Gradually decrease SF when stable link quality is observed.

Although ERAC uses point-to-point LoRa without LoRaWAN ADR, this custom responsiveness significantly improves PDR during wave occlusion periods.

6.5.2 Congestion Control and Collision Avoidance

To prevent congestion collapse during a debris event (where multiple nodes might detect debris simultaneously), ERAC implements a Probabilistic Transmission Delay. Upon detecting debris, a node does not transmit immediately ($t = 0$). Instead, it waits a random time $t \in$, where $T_{contention}$ is inversely proportional to the node's calculated confidence in the detection. This desynchronizes the burst of alerts from neighboring nodes.

ERAC also uses:

- priority-weighted backoff
- energy-aware retransmission
- dynamic duty-cycle throttling during congestion

These mechanisms prevent collision cascades and ensure high-priority alerts reach the master reliably even in dense event-driven bursts.

6.6 Simulation Environment Setup (NS-3/AquaSim):

High-fidelity simulation is essential to evaluate the scalability, reliability, and energy performance of ERAC under controlled yet realistic marine conditions. NS-3 is selected due to its modular architecture, robust PHY/MAC modeling capabilities, and extensibility for custom protocol development.

Simulation Platform:

- NS-3 (C++): Used to implement ERAC’s MAC logic, energy model, and mobility model.
- Marine Channel Extensions (AquaSim-NG components): Used selectively to approximate near-surface fading, wave-induced occlusion, and mobility effects.

Implemented Modules:

- Custom LoRa PHY/MAC Module: A lightweight LoRa PHY (SF, bandwidth, Tx power) is implemented without the constraints of LoRaWAN Class A. This allows ERAC to control timing, duty cycle, and backoff behavior directly.
- Dual-Source Energy Harvester Model: Solar and wave energy inputs are modeled via synthetic, stochastic traces generated from EWMA-based (solar) and IMU-like (wave) profiles.
- Battery Model: A Li-Ion State-of-Charge (SoC) model capturing discharge rate, charging efficiency, and voltage–capacity relationship.
- Drift Mobility Model: Simulates buoy drift influenced by simplified current vectors and random wave motion.

Simulation Scenarios: Two primary scenarios are used.

Scenario A – Baseline (Non-Adaptive LoRa MAC):

- Standard ALOHA-like uplink scheduling
- No energy prediction
- No QoS differentiation
- Used only as a comparative baseline

Scenario B – ERAC Enabled:

- Full ERAC MAC operation
- Predictive duty cycling
- Fast-drop/slow-rise SF adaptation
- Priority queue scheduling
- Probabilistic congestion control

Evaluation Metrics:

- Packet Delivery Ratio (PDR):
Measured separately for high-priority and low-priority packets.
- Latency (High-Priority Traffic):
End-to-end delay under event-driven bursts.
- Energy Neutrality:
Whether each node maintains $\int P_{harv} \geq \int P_{cons}$ over a 7-day simulated period.
- Network Lifetime:
Time until first node dies in the baseline vs. ERAC scenario.

These simulations quantify the advantages of ERAC over non-adaptive MAC approaches in realistic marine settings.

6.7 Hardware Implementation Methodology

Hardware implementation validates ERAC's practical operability under real-world energy and communication constraints. A scaled-down prototype is developed to test firmware logic, physical energy measurements, and LoRa performance.

- Hardware Stack:
 - Microcontroller: e.g. - ESP32-S3/ Raspberry Pi (selected for low power and extensive library support).
 - Transceiver: e.g. - Semtech SX1276/SX1262 (LoRa module).
 - Energy: e.g. - 6V Solar Panel, 18650 Li-Ion Battery, INA219/INA3221 Current Sensor (for measuring harvesting/consumption). Wave energy input will be simulated via a variable voltage source or shaker table if available.
- Firmware: Developed in C++/Arduino framework. It implements the "Lightweight ERAC" – including:
 - solar EWMA prediction
 - IMU-based wave-energy estimation (using synthetic IMU patterns for testing)
 - dynamic duty cycling
 - multi-queue priority scheduler
 - adaptive SF logic
 - energy-aware backoff

Nodes communicate using direct LoRa packets with custom MAC headers (priority, queue status, predicted energy).

- Testing Procedure:

1. Dry Lab Evaluation:

- Solar panel partially covered/uncovered to simulate fluctuating harvesting.
- Duty-cycle response logged and compared against predicted energy.
- Verification of priority queue transitions under synthetic debris events.

2. Short-Range Outdoor Test:

- Sub-nodes placed at varying distances.
- Test robustness under obstruction (people walking between nodes, partial shading).
- Validate fast-drop/slow-rise SF adjustments.

3. Water-Tank / Pond Test (if available):

- Test surface reflection/occlusion effects.
- Evaluate packet losses and recovery behavior under mild wave action.

6.8 Tasks and Timeline:

The project spans approximately 10 months (Aug 2025 – Jun 2026) and is divided into five sequential phases to ensure structured progress and thorough validation of the ERAC protocol.

Table 1: Tasks and Timeline

Phase	Focus	Key Activities	Expected Deliverables	Duration
Phase I	Research & Design	Survey EH-MAC, QoS; Develop ERAC architecture & math models	ERAC Specification Doc	Aug – Oct 2025
Phase II	Implementation	Implement Custom NS-3 Modules (Energy, MAC); Firmware Dev	Functional NS-3 Prototype	Nov 2025 – Jan 2026
Phase III	Hardware Prototype	Build Microcontroller + LoRa Nodes; Integrate Sensors/Energy	Functional Hardware Nodes	Jan – Mar 2026
Phase IV	Validation	Execute Simulations; Conduct Field Tests; Analyze Data	Simulation Results & Data	Mar – May 2026
Phase V	Reporting	Final Thesis Writing; Demonstration Preparation	Final Report & Demo	May – Jun 2026

Note: A detailed Gantt Chart depicting activity overlaps and time allocation is provided in Appendix A for reference.

7 PROJECT REQUIREMENTS

The success of the ERAC protocol is evaluated based on its ability to meet functional expectations and strict performance targets.

7.1 Functional Requirements:

- **FR-1 (Energy Monitoring):** The system must continuously monitor E_{res} (Battery Voltage) and P_{harv} (Charging Current) with a sampling interval no greater than 1 minute.
- **FR-2 (Hybrid Energy Forecasting):** The system shall continuously collect solar and wave-energy measurements and compute short-term predictions using EWMA (solar) and IMU-derived variance models (wave).
- **FR-3 (Dynamic Duty Cycle Adaptation):** The MAC shall adjust the node's sleep-wake duty cycle based on predicted energy availability to ensure Energy Neutral Operation (ENO).
- **FR-4 (Priority-Based Packet Scheduling):** The MAC shall classify outgoing packets into at least two priority levels (Critical and Routine) and schedule transmissions accordingly.
- **FR-5 (Adaptive LoRa Transmission Parameters):** The system shall dynamically adjust Spreading Factor (SF) and Transmission Power based on link quality trends and packet priority.
- **FR-6 (Energy-Aware Transmission Decision):** Before transmitting any packet, the system shall verify that residual energy remains above the required reserve threshold for the next operational cycle.
- **FR-7 (Congestion Mitigation & Collision Avoidance):** The MAC shall implement probabilistic backoff and priority-weighted delays to reduce collisions during event-driven bursts.
- **FR-8 (Connectivity Monitoring & Link Recovery):** The system shall monitor signal strength or beacon reception and adjust SF upward when connectivity degrades ("fast-drop, slow-rise" behavior).
- **FR-9 (Point-to-Point LoRa Communication):** Nodes shall communicate with the master node using direct LoRa packets with custom MAC headers, not LoRaWAN procedures.
- **FR-10 (Data Forwarding by Master Node):** The master node shall collect packets, verify MAC headers, and forward validated data to the cloud backend via LTE/5G or satellite.

7.2 Non-Functional Requirements:

The following non-functional requirements define the necessary performance standards.

Table 2: Quantifiable Non-Functional Requirements (Target Metrics)

Requirement Type	Metric	Baseline Target (Non-Adaptive MAC)	ERAC Target Improvement	Justification
Energy Efficiency	Network Lifetime Extension (L_{total})	Baseline LoRa MAC Lifetime (L)	L + 30%	Achieves long-term ENO using predictive solar + wave energy management
Reliability	Packet Delivery Ratio (PDR)	75 - 80% (Surface marine mobility)	> 90% (Critical Data)	ERAC prioritizes critical alerts and adapts SF quickly under link degradation
Latency (Critical Data)	End-to-End Delay (High Priority Traffic)	$T_{baseline}$	Reduced by 25%	Necessary for timely debris-event reporting
Resilience	Recovery Time	No resilience mechanism	< 1 min (Link Recovery)	Fast-drop/slow-rise SF logic allows recovery after wave occlusion
Energy Awareness	Prediction Accuracy	No forecasting	Maintains ENO over 7-day cycle	Forecast-guided duty cycling prevents depletion

7.3 System and User Requirements:

System Requirements (SR)

The system requirements define the essential platform, architectural, and component capabilities necessary to implement and validate the ERAC protocol:

- **SR-1 (Platform):** ERAC must be developed and validated using NS-3 with marine fading and drift mobility components.
- **SR-2 (Language):** ERAC MAC logic, energy models, and adaptive scheduling must be implemented in C++.
- **SR-3 (Hardware):** The hardware prototype uses ESP32/ Raspberry Pi microcontrollers, SX1276/SX1262 LoRa transceivers, and solar + simulated wave-energy input.
- **SR-4 (Energy Monitoring):** The platform shall support real-time measurement of battery voltage and charging current.
- **SR-5 (MAC Processing):** The firmware must support priority queues, adaptive SF control, and energy-aware duty cycling.

User Requirements (UR)

The user requirements define the expected operational performance from the perspective of marine monitoring decision-makers and system maintainers:

- **UR-1 (Critical Latency):** High-priority alerts shall meet strict latency constraints (NFR: 25% delay reduction).
- **UR-2 (Continuous Operation):** Nodes shall maximize operational lifetime using energy harvesting and adaptive duty cycling (NFR: $L_{total} + 30\%$).
- **UR-3 (Data Integrity):** Critical packets must achieve high reliability despite channel impairments (NFR: PDR > 90%).
- **UR-4 (Resilience):** The communication system must maintain connectivity despite wave-based occlusion and mobility.
- **UR-5 (Differentiated Service):** High-priority event data must be processed before routine telemetry under all energy states.

7.4 Use/Test Cases (Tentative):

Validation relies on large-scale simulation testing and small-scale hardware demonstration, ensuring ERAC satisfies the defined functional and non-functional requirements.

Four key test cases are designed to isolate and verify the core functional and non-functional requirements of the ERAC protocol:

Table 3: Demonstration Focused Test Cases

Test Case ID	Scenario Description	Variables Tested	NFR Validation	ERAC Modules Under Test
TC-MAC-01	Stochastic Energy Profile Simulation (Variable solar/wave input over 48h).	Duty Cycle Adaptation Rate, Energy Buffer Level, Forecast Tracking	Energy Efficiency	Energy Forecasting, Adaptive Duty Cycle
TC-QoS-02	High-Priority Data Burst (Multiple nodes detect debris simultaneously).	Latency, PDR for critical data, Queue Stability	Latency, Reliability	QoS-Aware Scheduler, Cross-Layer Interface
TC-RES-03	Link Failure/Occlusion (Simulated wave blocking signal).	Recovery time, SF adjustment speed.	Reliability, Resilience	Adaptive Link Maintenance (Fast-drop/Slow-rise Logic)
TC-HW-04	Hardware Energy Awareness (Physical solar panel covered/uncovered).	Duty Cycle Reaction Time, Battery Drain Profile	Energy Efficiency	Hardware Firmware Implementation

8 BUDGET AND BUDGET JUSTIFICATION

The development and validation of the Energy-Resilient Adaptive Communication (ERAC) protocol primarily leverage open-source simulation tools, existing laboratory infrastructure, and shared hardware resources within the OceanSense project. As a result, the proposed work does not require a separate financial budget specifically for this contribution. All essential physical components used for the hardware prototype fall under the hardware design allocation managed by the respective team member responsible for the node architecture.

Budget:

There is no direct monetary budget allocated solely for the ERAC protocol implementation.

All required hardware components, including microcontrollers, LoRa transceivers, solar panels, current sensors, and energy storage elements are already provisioned under the hardware development contribution of another team member.

Software tools utilized in the ERAC workflow include:

- NS-3 (Network Simulator 3): Open-source
- AquaSim-NG Extensions: Open-source
- Arduino IDE: Free software
- Python/Matlab: Available through institute licenses (as applicable)

Therefore, no additional expenditure is requested for simulation, firmware development, or testing.

Budget Justification:

The project design strategically maximizes the use of open-source frameworks and existing laboratory equipment, enabling comprehensive validation of ERAC without additional budget requirements. The justification is detailed as follows:

1. Simulation Cost Efficiency

NS-3, together with custom C++ modules and marine propagation models, provides a robust platform for validating ERAC's energy, latency, and reliability performance at zero cost.

2. Shared Hardware Resources

The physical prototyping phase uses components already acquired for the hardware design contribution. These include:

- Microcontrollers

- SX1276/SX1262 LoRa modules
- Solar harvesting units
- INA219/INA3221 current sensors
- Li-Ion cells and supporting circuitry

This ensures the ERAC firmware can be tested on actual embedded hardware without incurring new expenses.

3. Institutional Support

Laboratory test environments—such as controlled lighting rigs, water-tank setups, and measurement tools—are available as shared departmental resources.

4. Zero-Cost Development Stack

All programming, debugging, and analysis tools adhere to zero-cost or institution-licensed software, eliminating the need for dedicated financial allocation.

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APPENDICES

Appendix A: Project Gantt Chart

